airtel

SYNC

Where **Ingenuity**
Meets Scale.

Case Study



Platform Partner unstop

NETWORK CASE STUDY · CASE 1

Stadium Surge

5G capacity for a 1,00,000-fan final

The situation

A cricket World Cup final at a 1,00,000-seat stadium is 60 days away. At the last big match, with 65,000 fans, the network choked at the innings break. 4G load hit about 99%, and user speeds fell from about 16 Mbps to 1 Mbps. Most 5G phones could not connect to 5G, and UPI payment failures at food stalls rose from 1% to 7%.

60 days	**~99%**	**16 → 1 Mbps**	**1% → 7%**
to the final	4G load at innings break	user speed drop	UPI payment failures

> You have **₹4 crore** to make sure it doesn't happen again.

Key numbers

Key number	Value
Airtel users in the crowd	40% of fans
Share of them with 5G phones	70%
Share online at the same moment (peak)	18%
Target speed per user	5 Mbps on 5G · 3 Mbps on 4G
Spectrum	5G: 100 MHz (3.5 GHz) · 4G: 70 MHz across 3 bands
Costs	Cell on wheels ₹25 lakh per event · Small cell ₹8 lakh · Fiber backhaul ₹50 lakh per km

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Your tasks

1. Estimate peak demand and how many extra 5G and 4G cells the stadium needs @ 80% load, maximising the spectral efficiency.
2. Propose the mix of cells on wheels, small cells and upgrades that fits within ₹4 crore.
3. Explain how you would stop 5G phones getting stuck on 4G, and how you'd protect payments and emergency calls.
4. Do you anticipate any impact on user experience after deploying the proposed solutions? If yes, how can it be mitigated?

Data provided

> Sheet **sync** has load and user speed for the 8 stadium sites at four match moments (normal day vs last match), the site list with backhaul, and UPI failure rates.

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